

G. ARAVIND

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EXPERIENCE

Microsoft

Aug 2022 - Present

Data & Applied Scientist (Responsible AI)

Bengaluru, India

- Focusing on trust, fairness, responsible use of AI/LLMs, and preventing misinformation, hate speech, and unsafe content from appearing in Bing and Copilot.
- Trained transformer model with inference optimization to identify harmful content on long multi-turn conversations with Copilot, and implemented guardrails to block problematic responses in real time.
- Trained low-latency document classifier to filter pages from Bing's Web Index and Copilot grounding (40B+ URLs) that contain bad content, reducing average latency by 64% while maintaining same offline metrics.
- Designed and implemented metrics and evals for detecting unsafe content and IP/Copyright violations in chat conversations, for offline evaluation by 30+ partner teams.
- Simulated adversarial conversations using LLMs for various human personas to evaluate Copilot's safety.
- Piloted prompt-based routing of potentially problematic queries to "safer" responding models in Copilot.

Data & Applied Scientist (Bing Segments)

Hyderabad, India

- Shipped multiple developer-centric answers to production on Bing SERP, increasing feature consumed DAU by over 500K in 6 months.
- I worked on classifier models, knowledge graphs, personalisation, content extraction, answer triggering, and data refresh pipelines for the developer answers on Bing search.

Mastercard

Jun 2021 - Jul 2021

AI Specialist Intern

Gurgaon, India

- Trained models for predicting high-value card transactions and the time and type of future transactions, by modeling the transaction sequences as Temporal Point Processes.

EDUCATION

Indian Institute of Science, Bengaluru

Aug 2020 - Jul 2022

M.Tech in Artificial Intelligence

Grade: 8.0 / 10

Heritage Institute of Technology, Kolkata

Aug 2016 - Jul 2020

B.Tech in Computer Science & Engineering

Grade: 8.8 / 10

PROJECT

Adversarial Attacks on Recommendation System Models

- Modeled recommenders as a user-item bipartite graph, and developed new gradient-based attacks that optimally perturb the graph structure (adding new malicious users or user-item interactions).

TECHNICAL SKILLS

Programming Languages: Python, SQL, C#

Tools & Frameworks: Pytorch, Tensorflow, Semantic Kernel, Langchain, Scikit-learn, Azure ML, AWS, GCP, Open AI, Cosmos DB, Numpy, Pandas, Hugging Face, NLP, Search, Recommendation Systems, Git

OTHER

- All India Rank 6th in GATE Computer Science exam, out of ~1,00,000 candidates
- Teaching Assistant for E2 202: Random Processes at IISc, Fall 2021